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# Héctor Barreiro Cabrera

## Ph.D. in Computer Science

Passionate about technology and computing. I was born and raised in Valencia (Spain), but I currently live in Madrid, where I obtained my PhD in Computer Science under the guidance of Professor Miguel A. Otaduy. My love for computer graphics has led me to dedicate my academic and professional career to learn and research topics related to these fields: physics-based animation of deformable objects and fluids, tactile interaction with ultrasonic haptic devices, virtual and augmented reality, machine learning applied to simulation, high-performance parallel computing, real-time rendering techniques, ...

### PROFESSIONAL AND RESEARCH INTERESTS

- Physics-based animation
- Real-time rendering
- High-performance computing
- Machine learning
- Mixed reality (AR & VR)
- Haptic rendering

### PUBLICATIONS

Aug. 2021
**Soft-Tissue Simulation for Computational Planning of Orthognathic Surgery**
P. Alcañiz, J. Pérez, A. Gutiérrez, **H. Barreiro**, Á. Villalobos, D. Miraut, C. Illana, MA. Otaduy
Journal of Personalized Medicine

Jul. 2021
**Natural Tactile Interaction with Virtual Clay**
**H. Barreiro**, J. Torres, MA. Otaduy
Proc. of World Haptics Conference, 2021

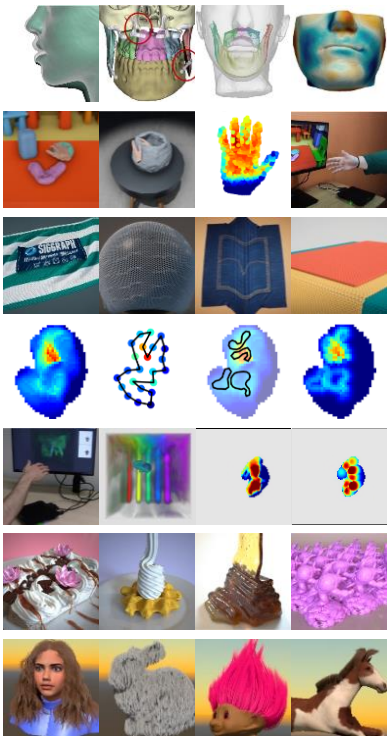
Jul. 2020
**Robust Eulerian-on-Lagrangian Rods**
R.M. Sánchez-Banderas, A. Rodríguez, **H. Barreiro**, MA. Otaduy
ACM Trans. on Graphics (Proc. of ACM SIGGRAPH), Volume 39, Number 4 - 2020

Feb. 2020
**Path Routing Optimization for STM Ultrasound Rendering**
**H. Barreiro**, S. Sinclair, MA. Otaduy
IEEE Trans Haptics. 2020 Feb 24. doi: 10.1109/TOH.2019.2963647.

Jul. 2019
**Ultrasound Rendering of Tactile Interaction with Fluids**
**H. Barreiro**, S. Sinclair, MA. Otaduy
2019 IEEE World Haptics Conference (WHC). IEEE, 2019

Nov. 2017
**Conformation Constraints for Efficient Viscoelastic Fluid Simulation**
**H. Barreiro**, I. García-Fernández, I. Alduán, MA. Otaduy
ACM Trans. on Graphics (Proc. of ACM SIGGRAPH Asia), 2017

Jul. 2015
**Real-time Inextensible Hair with Volume and Shape**
R. M. Sánchez-Banderas, **H. Barreiro**, I. García-Fernández, M. Pérez Martínez
Congreso Español de Informática Gráfica, 2015



## EDUCATION

### Ph.D. in Computer Science
Higher School of Computer Engineering, Universidad Rey Juan Carlos, Spain
Sep. 2016 – Sep. 2021
Thesis focused on the development of novel techniques for simulating physical phenomena and haptic rendering.

### Master's Degree in Computer Graphics, Videogames and VR
Higher School of Computer Engineering, Universidad Rey Juan Carlos, Spain
Sep. 2015 – Jul. 2016
Master's degree in computer graphics covering diverse subjects such as rendering techniques, graphic processors, and physics-based simulation, as well as videogames and virtual reality.

### Bachelor's Degree in Multimedia Engineering
Higher School of Engineering, Universitat de València, Spain
Sep. 2010 – Sep. 2015
Degree that combines audiovisual communication with computer engineering, especially deepening in multimedia systems and all related areas (graphics, simulation, sound, ...).

## COURSES & CERTIFICATES

### Machine Learning
Stanford Online. @ Coursera

### Neural Networks and Deep Learning
Deeplearning.ai @ Coursera

### Improving Deep Neural Networks: Hyperparameter tuning
Deeplearning.ai @ Coursera

## WORK EXPERIENCE

### Research Engineer
Meta Reality Labs Research, Redmond, USA
May 2021 – Apr 2022
R&D of simulation framework for AR/VR

### Research Intern
Meta Reality Labs Research, Redmond, USA
Aug. 2020 – Nov. 2020
Acceleration strategies for high-fidelity simulation methods

### Researcher
Universidad Rey Juan Carlos, Móstoles, Spain
Nov. 2015 – Apr. 2021
Development of constraint-based model for the simulation of extremely viscous and viscoelastic fluids. Development of haptic interaction models with virtual fluids.

### Research Intern
Ultraleap Ltd, Bristol, UK
Feb. 2020 – Mar. 2020
Integration of the Handybeam acoustic simulator with MSLab's soft-body simulation framework.

### Researcher
Next Limit Technologies, Madrid, Spain
Dec. 2017 – Sept. 2018
Research into the extension of fluid simulation methods and the use of machine learning techniques. Funded by Spain's government under the *Doctorados Industriales* program (*ref. DI-16-08640*).

### Junior Programmer
Instituto de robótica y tecnologías de la información y comunicación. Paterna, Spain
Oct. 2013 – Feb. 2015
Development and implementation of software for the simulation of port machinery and interactive applications

## PROFICIENCIES

Spanish: Native
English: B2*

* Without formal qualification, skill levels stipulated following the CEFR self-assessment table provided by the Instituto Cervantes.

## PROGRAMMING LANGUAGES

C++ C# Python

## TOOLS OF CHOICE

| | |
|---|---|
| IDE | Visual Studio, VS Code, Spyder, Jupyter |
| Frameworks | Eigen, Numpy, Scipy, Sympy |
| Source control | Git, Mercurial |
| HPC | CUDA, OpenCL, DirectCompute |
| Game engines | Unity |

**For further information, please contact me or visit my online portfolio.**

Thank you for your time